Practical - 5

Registration Form with Frontend Validation and User-Friendly Error Handling

Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

A. Theory

1. Are correct input types and attributes used?

Answer :

Yes. The forms utilize HTML5 input types appropriately, such as type="email" for email validation, type="password" for masking characters, and the required attribute to leverage built-in browser constraints.

2. Are validation errors displayed near the relevant fields?

Answer :

No. Validation errors are handled via browser popups using alert("Invalid Name! ..."). They are not dynamically rendered into the HTML DOM next to the specific fields that caused the error.

3. Is password strength checked?

Answer :

No. The JavaScript only checks if the password matches the confirmPassword. There are no Regular Expressions enforcing complexity rules (like requiring numbers, uppercase letters, or special characters).

4. Is the form accessible using keyboard and screen-reader-friendly labels?

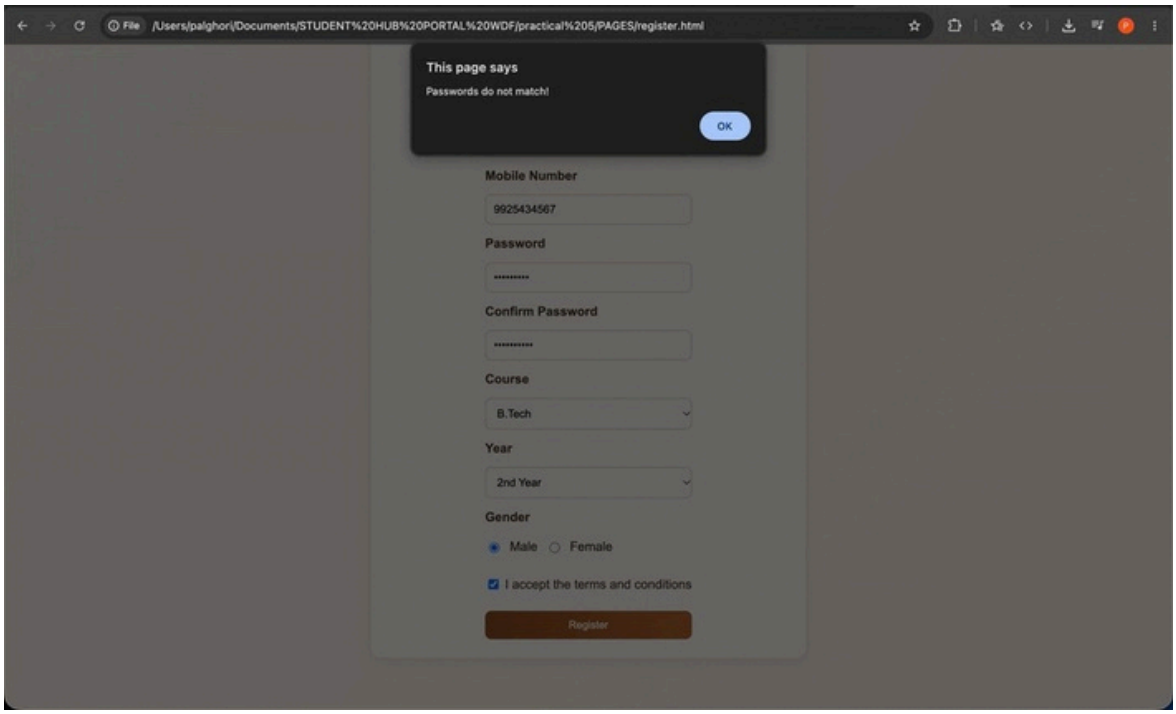
Answer :

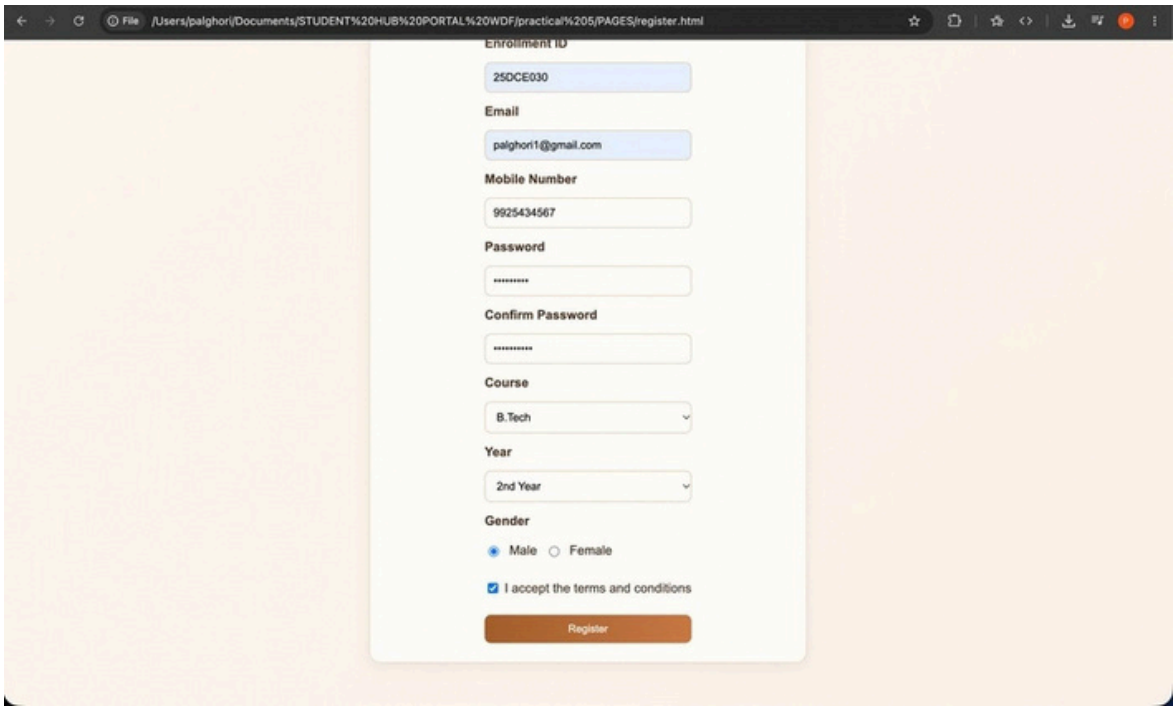
Mostly yes. The form uses <label> tags, making it somewhat screen-reader friendly, and inputs are easily navigable via the Tab key. However, not all labels use the for attribute (e.g., <label>Full Name</label>). To be strictly accessible, every label should have a for="id_of_input" attribute to explicitly link them.

B. Implementation

Expanded the JavaScript to handle comprehensive Client-Side form validation. Implemented complex Regular Expressions to validate multiple formats of Enrollment IDs (e.g., standard vs. D2D formats), strictly alphabetical Names, and exactly 10-digit Mobile numbers. Also added logic to ensure password and confirmPassword match.

C. Evidence of Work (Screenshots)




File
/Users/palghori/Documents/STUDENT%20HUB%20PORTAL%20WDF/practical%205/PAGES/register.html
Enrollment ID
25DCE030
Email
palghori1@gmail.com
Mobile Number
9925434567
Password
Confirm Password
Course
B.Tech
Year
2nd Year
Gender
Male
Female
I accept the terms and conditions
Register

D. GitHub Repository Link:

https://github.com/palghori/STUDENT-HUB-PORTAL-WDF/tree/main/practical%205

E. Challenges Faced:

- Writing and debugging robust Regular Expressions (Regex) that don't accidentally block valid data or allow incorrect formats (like /(^\d{2}[a-zA-Z]{3}\d{3}$)|(^D2D[a-zA-Z]{2,3}\d{3}$)/i).
- Capturing form submission events and safely halting them (event.preventDefault()) so that the user receives an alert() error instead of the page reloading and losing their typed data.

F. Learning Outcomes:

Gained a deep understanding of Regex pattern matching, securing forms on the frontend, and improving User Experience (UX) by providing immediate feedback on invalid inputs before the server is ever contacted.

Faculty Remarks: